Hopfield Network: Recalling Noisy Letters

A small project where I built a Hopfield network from scratch using only NumPy and used it to restore noisy letters (C, H, T, X) drawn on a 7x5 pixel grid.

What's inside

- `Hopfield Network.ipynb` - the full notebook (training, recall, energy plot, noise test, capacity test)

What I did

1. Drew letters as +1 / -1 pixel grids
2. Trained the weights with the Hebbian rule
3. Recalled patterns from noisy inputs using asynchronous updates
4. Plotted the energy to see the network settling into a stable state
5. Tested accuracy at different noise levels
6. Tested how many patterns the network can store (about 0.14 x N)
7. Showed that similar letters (A/H, C/L) interfere with each other

Results

- 4 letters: about 93% exact recall at 15% noise, and 100% with no noise
- Recall drops sharply once 40%+ of pixels are flipped
- Storing all 6 letters breaks recall for H and L

How to run

```
pip install numpy matplotlib jupyter
jupyter notebook
```

Open the notebook and run all cells.

Tech

Python, NumPy, Matplotlib